ARM PrimeCell

Advanced Audio Codec Interface (PL041)

Release Note

PL041 RELN 0000 B

ARM PrimeCell Advanced Audio CODEC Interface (PL041) Release Note

Proprietary notice

Document confidentiality status

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Product status

The information in this document is Preliminary (information on a product in development).

Web address

http://www.arm.com

Feedback

ARM limited welcomes feedback on both the product, and the documentation.

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General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell Advanced Audio Codec Interface

If you have any comments or suggestions about this product, please contact your supplier giving:

- The Product name
- A concise explanation of your comments.

Change history

Issue	Date	Change
A	5th September 2000	First release.
B	16th July 2001	Software driver files are modified

Contents

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL041 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL041-MN-23001-REL1v0
Synthesizable Verilog RTL	N/A	PL041-MN-22001-REL1v0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB25 cell library)	N/A	PL041-MN-01002-REL1v0
Testbench – Functional Verification, VHDL Testbench – EASY Integration test, VHDL	N/A N/A	PL041-MN-23010-REL1v0 PL041-MN-23012-REL1v0
Testbench – Functional Verification, Verilog	N/A	PL041-MN-22010-REL1v0
Testbench – EASY Integration test, Verilog	N/A	PL041-MN-22012-REL1v0
Test vectors-BusTalk, functional verification	N/A	PL041-VE-01010-REL1v0
Test vectors-BusTalk, integration test	N/A	PL041-VE-01012-REL1v0
ARM PrimeCell Advanced Audio Codec Interface (PL041) Technical Reference Manual (PDF)	ARM DDI 0173 B	PL041-DA-03001-REL1v0
ARM PrimeCell Advanced Audio Codec Interface (PL041) Design Manual (PDF)	PL041 DDES 0000 A	PL041-DC-02002-REL1v0
ARM PrimeCell Advanced Audio Codec Interface (PL041) Integration Manual (PDF)	PL041 INTM 0000 A	PL041-DC-10002-REL1v0
ARM PrimeCell Advanced Audio Codec Interface (PL041) Release Note (PDF)	PL041 RELN 0000 B	PL041-DC-06002-REL1v0
System Common Source Code	N/A	PL000-SW-02001-REL2v0
System Indirection Source Code	N/A	PL000-SW-02003-REL2v0
System Test Source Code	N/A	PL000-SW-02004-REL2v0
System Build source code	N/A	PL000-SW-02005-REL2v0
Interrupt Controller Object Code	N/A	PL001-SW-00000-REL2v0
Timer Object Code	N/A	PL002-SW-02001-REL2v0
Software Code Module	N/A	PL041-SW-02001-REL2v0
System Structure and Integration Guide	PL000_DII_1000	PL000-DII-10000-REL2v0
System Common API Documents	PL000_ISPC_1001	PL000-DC-02001-REL2v0
System Indirection API Documents	PL000_ISPC_1002	PL000-DC-02001-REL2v0
System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-REL2v0

System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-REL2v0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-REL2v0
Software Test Report	PL041_TREP_1000	PL041-DC-01001-REL2v0
Software API Document	PL041_ISPC_1000	PL041-DC-02010-REL2v0

Table 1.1-1: ARM part numbers for ARM PrimeCell AACI PL041

Note Only the required deliverables will be supplied, as per the contractual agreement.

For each of the above, two files are generated using the UNIX tar create and the non-document tar files are then UNIX compress'ed. The files are named for each unique product revision, and the generic naming convention is illustrated below :

```
PL041-AA-NNNNN-REL1v0.tar.Z     - compressed tar file

PL041-AA-NNNNN-REL1v0.lst       - directory and file listing of above tar file
```

Where `-AA-NNNNN` represents the alpha-numeric product sub-type,
and REL1v0 represents the numeric release version of this peripheral.

1.2 File deliverables

This ARM PrimeCell peripheral consists of the following compressed 'tar' and 'list' file deliverables:

VC bundle

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/PL041-DC-06002-REL1v0.pdf

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/DAtasheet/PL041-DA-03001-REL1v0/PL041-DA-03001-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/DAtasheet/PL041-DA-03001-REL1v0/PL041-DA-03001-REL1v0.tar

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/DoCumentation/PL041-DC-02002-REL1v0/PL041-DC-02002-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/DoCumentation/PL041-DC-02002-REL1v0/PL041-DC-02002-REL1v0.tar

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/DoCumentation/PL041-DC-10002-REL1v0/PL041-DC-10002-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/DoCumentation/PL041-DC-10002-REL1v0/PL041-DC-10002-REL1v0.tar

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-01002-REL1v0/PL041-MN-01002-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-01002-REL1v0/PL041-MN-01002-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-22001-REL1v0/PL041-MN-22001-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-22001-REL1v0/PL041-MN-22001-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-22010-REL1v0/PL041-MN-22010-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-22010-REL1v0/PL041-MN-22010-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-22012-REL1v0/PL041-MN-22012-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-22012-REL1v0/PL041-MN-22012-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-23001-REL1v0/PL041-MN-23001-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-23001-REL1v0/PL041-MN-23001-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-23010-REL1v0/PL041-MN-23010-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-23010-REL1v0/PL041-MN-23010-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-23012-REL1v0/PL041-MN-23012-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/MN_native_model/PL041-MN-23012-REL1v0/PL041-MN-23012-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/VEctors/PL041-VE-01010-REL1v0/PL041-VE-01010-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/VEctors/PL041-VE-01010-REL1v0/PL041-VE-01010-REL1v0.tar.Z

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/VEctors/PL041-VE-01012-REL1v0/PL041-VE-01012-REL1v0.lst
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-01000-REL1v0/VEctors/PL041-VE-01012-REL1v0/PL041-VE-01012-REL1v0.tar.Z

Peripheral specific driver bundle

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-02000-REL2v0/Software/PL041-SW-02001-REL2v0/PL041-SW-02001-REL2v0.tar.Z
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-02000-REL2v0/Software/PL041-SW-02001-REL2v0/PL041-SW-02001-REL2v0.lst

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-02000-REL2v0/DoCumentation/PL041-DC-01001-REL2v0/PL041-DC-01001-REL2v0.tar
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-02000-REL2v0/DoCumentation/PL041-DC-01001-REL2v0/PL041-DC-01001-REL2v0.lst

PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-02000-REL2v0/DoCumentation/PL041-DC-02001-REL2v0/PL041-DC-02001-REL2v0.tar
PLXXX_PrimeCell/PL041-BU-00000-REL1v0/PL041-BU-02000-REL2v0/DoCumentation/PL041-DC-02001-REL2v0/PL041-DC-02001-REL2v0.lst

Common driver bundle

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02001-REL2v0/PL000-SW-02001-REL2v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02001-REL2v0/PL000-SW-02001-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02003-REL2v0/PL000-SW-02003-REL2v0.tar.Z

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02003-REL2v0/PL000-SW-02003-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02004-REL2v0/PL000-SW-02004-REL2v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02004-REL2v0/PL000-SW-02004-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02005-REL2v0/PL000-SW-02005-REL2v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL000-SW-02005-REL2v0/PL000-SW-02005-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL001-SW-00000-REL2v0/PL001-SW-00000-REL2v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL001-SW-00000-REL2v0/PL001-SW-00000-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL002-SW-02001-REL2v0/PL002-SW-02001-REL2v0.tar.Z
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/Software/PL002-SW-02001-REL2v0/PL002-SW-02001-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DC-02001-REL2v0/PL000-DC-02001-REL2v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DC-02001-REL2v0/PL000-DC-02001-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DC-02003-REL2v0/PL000-DC-02003-REL2v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DC-02003-REL2v0/PL000-DC-02003-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DC-02004-REL2v0/PL000-DC-02004-REL2v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DC-02004-REL2v0/PL000-DC-02004-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DII-10000-REL2v0/PL000-DII-10000-REL2v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL000-DII-10000-REL2v0/PL000-DII-10000-REL2v0.lst

PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL001-DC-02001-REL2v0/PL001-DC-02001-REL2v0.tar
PLXXX_PrimeCell/PL000-BU-00004-REL2v0/DoCumentation/PL001-DC-02001-REL2v0/PL001-DC-02001-REL2v0.lst

A generic peripheral user guide document (PDF) is supplied. This describes instructions for synthesis and simulation.
PLXXX_PrimeCell/PL000-BU-00003-REL2v0/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.lst
PLXXX_PrimeCell/PL000-BU-00003-REL2v0/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar

An install script is supplied which uncompresses and tar extracts (un-tar) the compressed tar files in order to build the peripheral directory structure.
PLXXX_PrimeCell/PL000-BU-00003-REL2v0/SoftWare/PL000-SW-99000-REL2v0/INSTALL

A reference cell library (Avant! CB25 0.25μm standard cells) is supplied with this peripheral.
PLXXX_PrimeCell/PL000-BU-00003-REL2v0/Avanti/CB25_1999.1.tar.Z
PLXXX_PrimeCell/PL000-BU-00003-REL2v0/Avanti/CB25_1999.1.lst

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX compressed tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 45 Mbytes of free disk space.

2.3 Installation procedure

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

1. **Relocate the shipment file**

   Relocate the encrypted file (for example <file>.tar.Z.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space.

2. **Decrypt file**

   Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tar.Z.

3. **Uncompress file**

   Execute the UNIX uncompress command to obtain the tar file <file>.tar:

```
uncompress <file>.tar.Z
```

   where <file> is the name of the file supplied.

Note Invoke gzip –d <file>.tar.gz, if gunzip compression has been applied.

4. **Extract tar files**

   To extract individual product tar files, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

   This will generate individual tar and list files for all of the products within this single shipment. Each peripheral consists of a set of tar files, and each tar file is named with the ARM part number that is listed in the contractual agreement document.

Note It is suggested that part numbers delivered are checked against the contractual agreement document.

2.3.2 Re-creating the peripheral directory structure

The following steps describe how to re-create the directory structure for each separate peripheral, which contains HDL RTL files, synthesis scripts, testbenches and documentation. These steps will need to be repeated for other peripherals in this shipment.

5. Choose method of installation

The directory structure can be extracted using the INSTALL script provided, or manually moving the files to the destination directory and using UNIX uncompress and tar commands. The INSTALL script moves the compressed tar files delivered, to a user specified destination directory and then uncompresses and tar extracts the files to re-create the peripheral directory structure. Manual installation may be preferred if only some of the deliverables are required.

Go to step 6 to use the INSTALL script provided, otherwise go to step 7 for manual installation.

All files begin with PLXXX, where XXX is a 3-digit value unique to each peripheral.

Note It is important to un-install or rename any version of this peripheral that already exists in the destination directory prior to uncompressing and extracting these tar files.

6. Run the install script

The install script supplied will automatically uncompress and tar extract files for the specified peripheral revision, in a user specified directory. When step 6 is complete there is then no need to perform steps 7 and 8, instead go to step 9.

Change to the directory containing the INSTALL script, for example (note, revision may be later than -REL1v0)

```
cd PLXXX_PrimeCell/PL000-BU-00003-REL2v0/SoftWare/PL000-SW-99000-REL2v0
```

To invoke the install C-shell script supplied, enter the following at the UNIX command line:

```
INSTALL <PRODUCT_NUMBER> <REVISION> <DESTINATION_DIRECTORY>
```

Options

<PRODUCT_NUMBER> = Each peripheral is defined by unique product code using the characters PL + 3-numeric digits. See below the shipment directory PLXXX_PrimeCell/

<REVISION> = Each peripheral has a revision number which has a status label and numeric version. See below the shipment directory PLXXX_PrimeCell/

<DESTINATION_DIRECTORY> = the full directory path where the peripheral is to be installed.

For example

A product shipped with the directory structure /PLXXX_PrimeCell/PL041-BU-00000-REL1v0/ has a product code PL041 and revision number Release REL1v0.

To install this peripheral to the the directory /ARM/PrimeCell_peripherals enter the following command line

```
INSTALL PL041 REL1v0 /ARM/PrimeCell_peripherals
```

This will install the peripheral having a product number PL041, with revision number REL1v0, to the destination directory /ARM/PrimeCell_peripherals/

```
Go to step 9
```

7. **Uncompress files**

   Copy all the compressed ‘tar’ files (*.tar.Z) for the peripheral, to the chosen destination directory. In the example below, the destination directory is called /ARM/PrimeCell_peripherals/.

   Use the UNIX uncompress command to obtain the tar files <file>.tar (* wildcards could also be used):

```
uncompress <file>.tar.Z
```

8. **Extract tar files**

   To extract each file individually, execute the UNIX tar extract command to build the peripheral file structure:

```
tar -xvf <file>.tar
```

9. **Peripheral directory installed**

   When all the peripheral files have been extracted, the destination directory (/ARM/PrimeCell_peripherals/) will contain:

   - the top-level peripheral directory (e.g. aaci_pl041) and underlying file structure of the deliverables, as well as the Release Note document file (PDF).
   - The remaining documentation for the peripheral (in Adobe Portable Document Format, *.pdf), is located in the aaci_pl041/docs/ directory.

2.3.3 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB25 1999.1 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PLXXX_PrimeCell/PL000-BU-00003-REL2v0/Avanti/CB25_1999.1.tar.Z .
```

Execute the UNIX uncompress and tar extract commands as below:

```
uncompress CB25_1999.1.tar.Z
tar -xvf CB25_1999.1.tar
```

2.3.4 Installation of the Generic User Guide

For instructions on performing simulations and synthesis, refer to the generic peripheral user guide.

The most recent revision of the generic user guide should be copied manually to a suitable location. An example is shown below to copy the generic user guide (revision REL1v0) to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PLXXX_PrimeCell/PL000-BU-00003-REL2v0/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar .
```

To extract the document file (PDF), execute the UNIX tar extract command:

```
tar -xvf PL000-DC-08001-REL1v0.tar
```

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at http://www.adobe.com. Select “Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell Advanced Audio Codec Interface PL041:
The documents files are located in the aaci_pl041/docs/ directory.

ARM PrimeCell Advanced Audio Codec Interface PL041
Technical Reference Manual ARM DDI 0173

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell Advanced Audio Codec Interface PL041
Integration Manual PL041 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell Advanced Audio Codec Interface PL041
Design Manual PL041 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release REL1v0, it has not been implemented in production silicon but has been proven on FPGA.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell Advanced Audio Codec Interface PL041.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 1999.1 CB25 v2.1 (0.25 μm)	Avant! Passport 1999.1 CB25 v2.1 (0.25 μm)
Total cell area, without scan, NAND-2 equivalents	26971.25	26660.25
Net interconnect area, without scan, NAND-2 equivalents	14516.33	14100.13
Total cell area, scan inserted,NAND-2 equivalents	33883.5	35968.5
Net interconnect area, scan inserted,NAND-2 equivalents	17352.94	16777.5
Clock frequency used	100 MHz	100 MHz
Scan insertion figure % (number of ‘D’ flip-flops replaced)	100% (2363)	100% (2363)
Estimated scan fault coverage, collapsed.	99.95%	99.94%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2000.05	Synopsys Design Compiler and Test Compiler 2000.05
Simulation tools and versions	Model Technology Inc, ModelSim v5.3d	Model Technology Inc, ModelSim v5.3d Cadence LDV 3.1
Global Peripheral Development Environment Version	Release REL1v1	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running HCLK.

4.3.1 Synchronous clocks

AACIBITCLK	PCLK	Simulations
12.5MHz (80 ns period)	100MHz (10 ns period)	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell Advanced Audio Codec Interface (PL041) Design Manual.

To change clock periods, modify following line in the verification/bustest/Aaci.h file:

```
#define  PCLK_PERIOD    10
#define  AACIBITCLK_PERIOD    10
```

And the /verification/$HDL_SOURCE/tbench/timing.vhd (or .v):

```
-- Define the low phase of clock
constant Tclkl       : time   := 5 ns;
-- Define the high phase of clock
constant Tclkh       : time   := 5 ns;
```

or in case of Verilog testbench:

```
// Clock frequency
`define  Tclkl    5
`define  Tclkh    5
```

5 DIFFERENCES FROM PREVIOUS REVISIONS

First release

6 KNOWN ISSUES

No known issues.

7 REVISION HISTORY

Date	Revision	Description
5th September 2000	REL1v0	Initial Release 1.0
21st June 2001	REL1v0	Software driver files are added
26th July 2001	REL1v0	Software driver files are modified